



Titanic Passengers Survival Data Analysis

```
In [ ]: # Import Libraries
```

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
```

```
In [ ]: # Load Dataset
```

```
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (8, 5)

df = pd.read_csv("train.csv")
print("Dataset shape:", df.shape)
df.head()
```

Dataset shape: (891, 12)

```
Out[ ]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cal
--	-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-----

0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	N
---	---	---	---	-------------------------	------	------	---	---	-----------	--------	---

1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
---	---	---	---	---	--------	------	---	---	----------	---------	---

2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	N
---	---	---	---	------------------------	--------	------	---	---	------------------	--------	---

3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C
---	---	---	---	--	--------	------	---	---	--------	---------	---

4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	N
---	---	---	---	--------------------------	------	------	---	---	--------	--------	---

```
In [ ]: print(df.info())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null   int64
1   Survived         891 non-null   int64
2   Pclass          891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp          891 non-null   int64
7   Parch          891 non-null   int64
8   Ticket         891 non-null   object
9   Fare          891 non-null   float64
10  Cabin         204 non-null   object
11  Embarked       889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None

```

```

In [ ]: # Summary statistics for numerical columns
print(df.describe())

```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```

In [ ]: # Summary statistics for categorical columns

print(df.describe(include=["O"]))

```

	Name	Sex	Ticket	Cabin	Embarked
count	891	891	891	204	889
unique	891	2	681	147	3
top	Dooley, Mr. Patrick	male	347082	G6	S
freq	1	577	7	4	644

```

In [ ]: print("\nValue Counts: Pclass")

```

```
print(df["Pclass"].value_counts())

print("\nValue Counts: Sex")
print(df["Sex"].value_counts())

print("\nValue Counts: Embarked")
print(df["Embarked"].value_counts())
```

```
Value Counts: Pclass
Pclass
3      491
1      216
2      184
Name: count, dtype: int64
```

```
Value Counts: Sex
Sex
male      577
female    314
Name: count, dtype: int64
```

```
Value Counts: Embarked
Embarked
S      644
C      168
Q       77
Name: count, dtype: int64
```

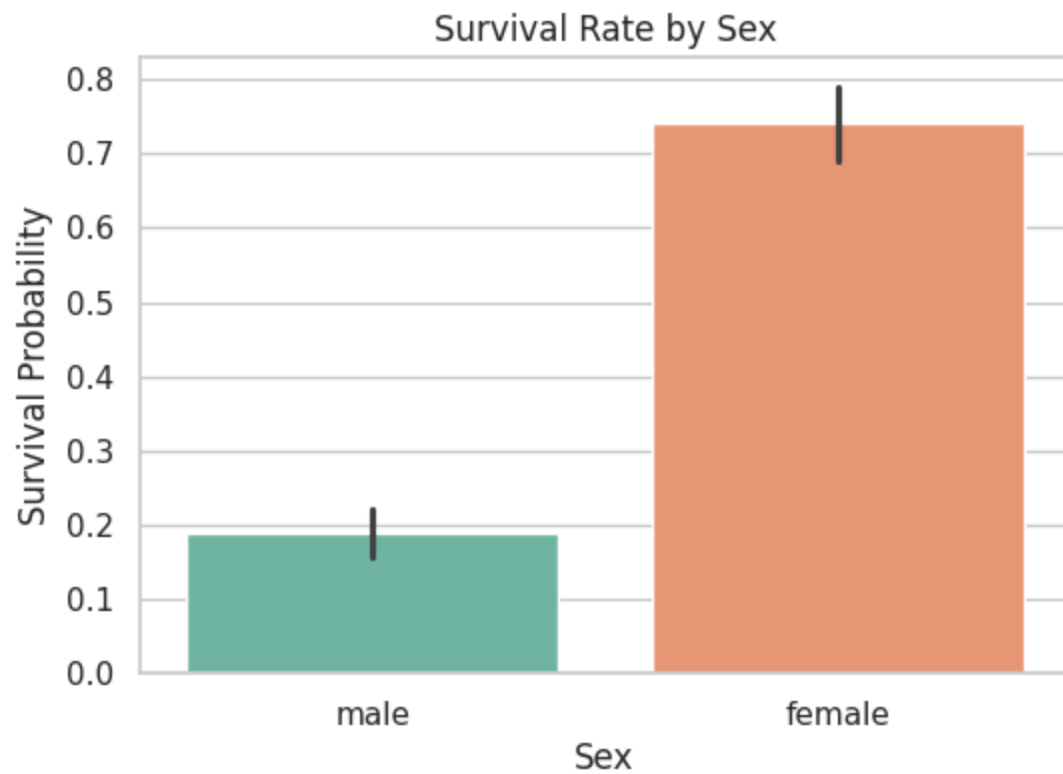
In []: *# Survival Rate by Sex*

```
plt.figure(figsize=(6, 4))
sns.barplot(x='Sex', y='Survived', data=df, palette='Set2')
plt.title('Survival Rate by Sex')
plt.ylabel('Survival Probability')
plt.show()
```

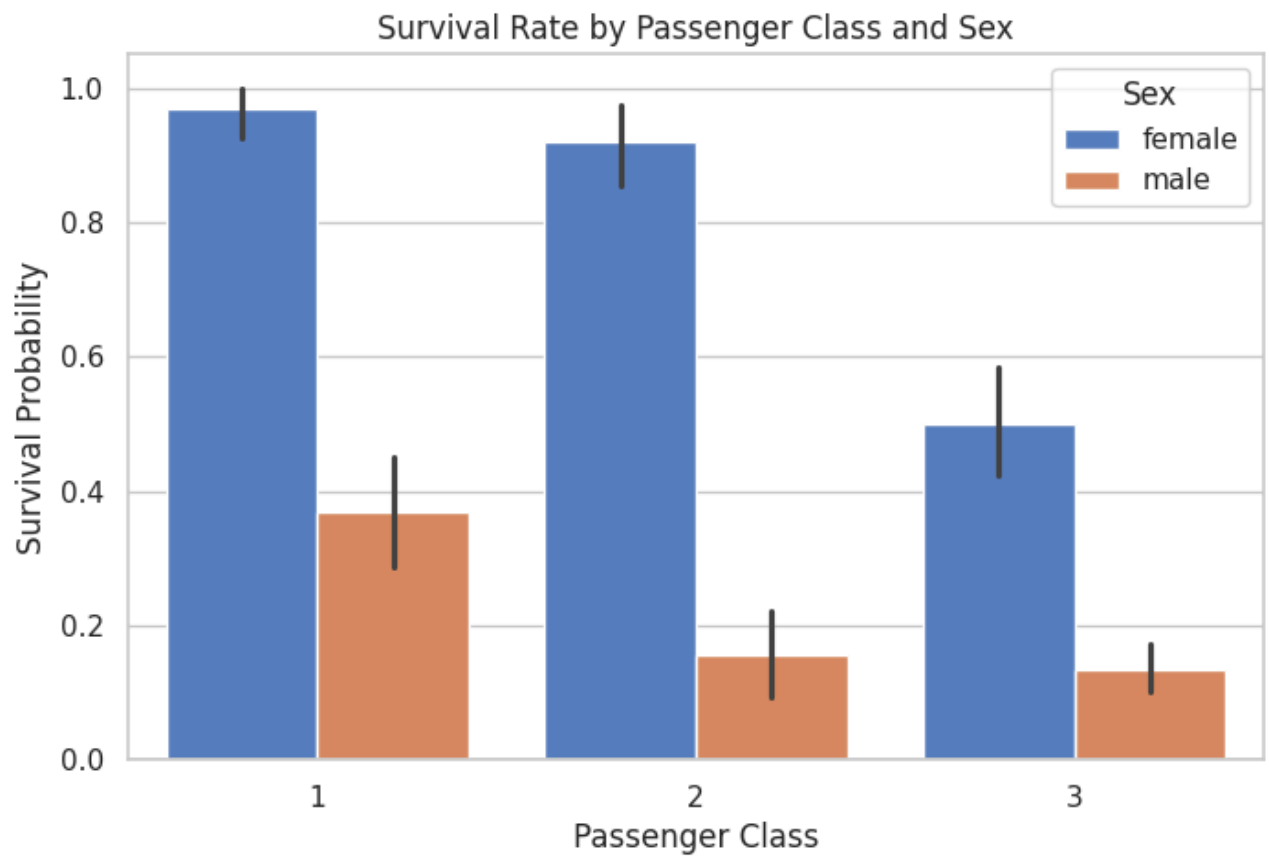
/tmp/ipykernel_840/3472111224.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

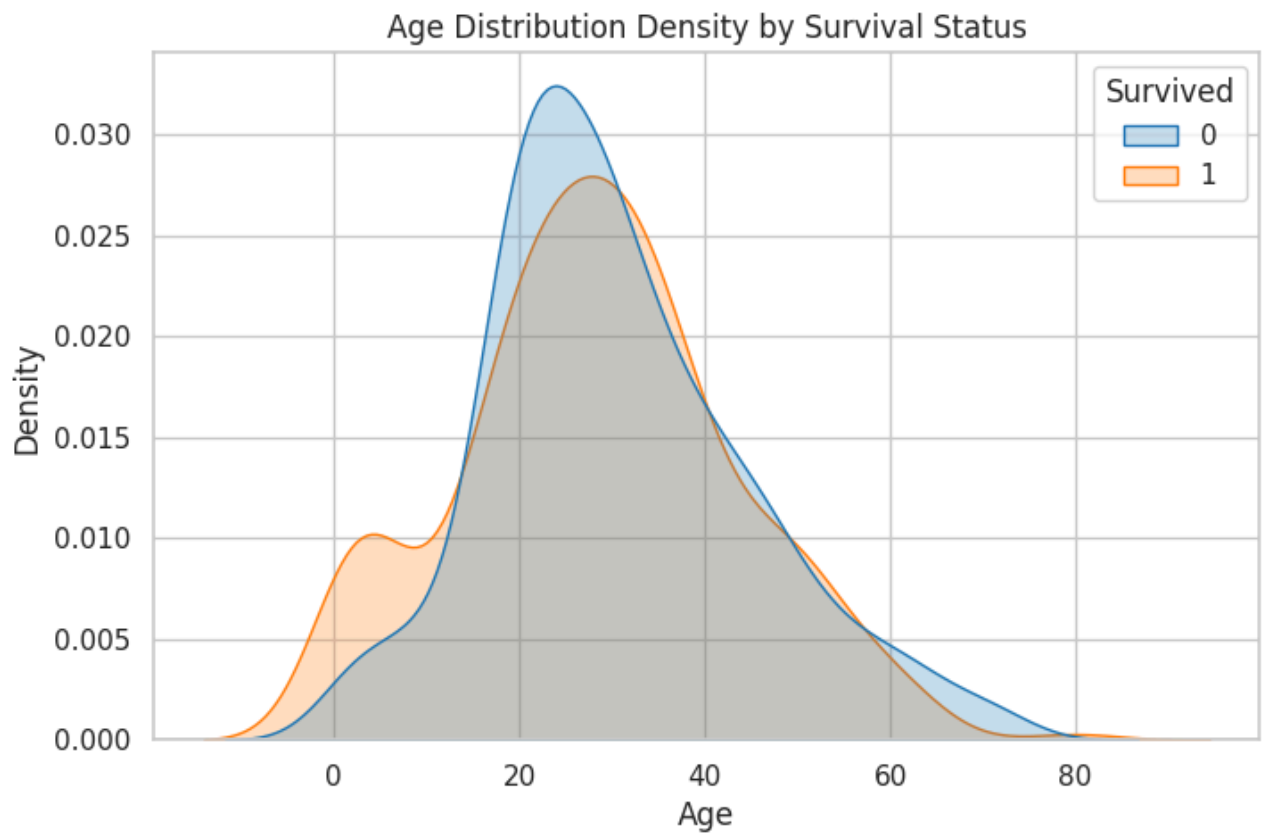
```
sns.barplot(x='Sex', y='Survived', data=df, palette='Set2')
```



```
In [ ]: # Survival Rate by Pclass and Sex
plt.figure(figsize=(8, 5))
sns.barplot(x='Pclass', y='Survived', hue='Sex', data=df, palette='muted')
plt.title('Survival Rate by Passenger Class and Sex')
plt.ylabel('Survival Probability')
plt.xlabel('Passenger Class')
plt.legend(title='Sex')
plt.show()
```



```
In [ ]: # Age Distribution by Survival Status
plt.figure(figsize=(8, 5))
sns.kdeplot(
    data=df, x='Age', hue='Survived', common_norm=False, fill=True, palette='magma'
)
plt.title('Age Distribution Density by Survival Status')
plt.xlabel('Age')
plt.show()
```



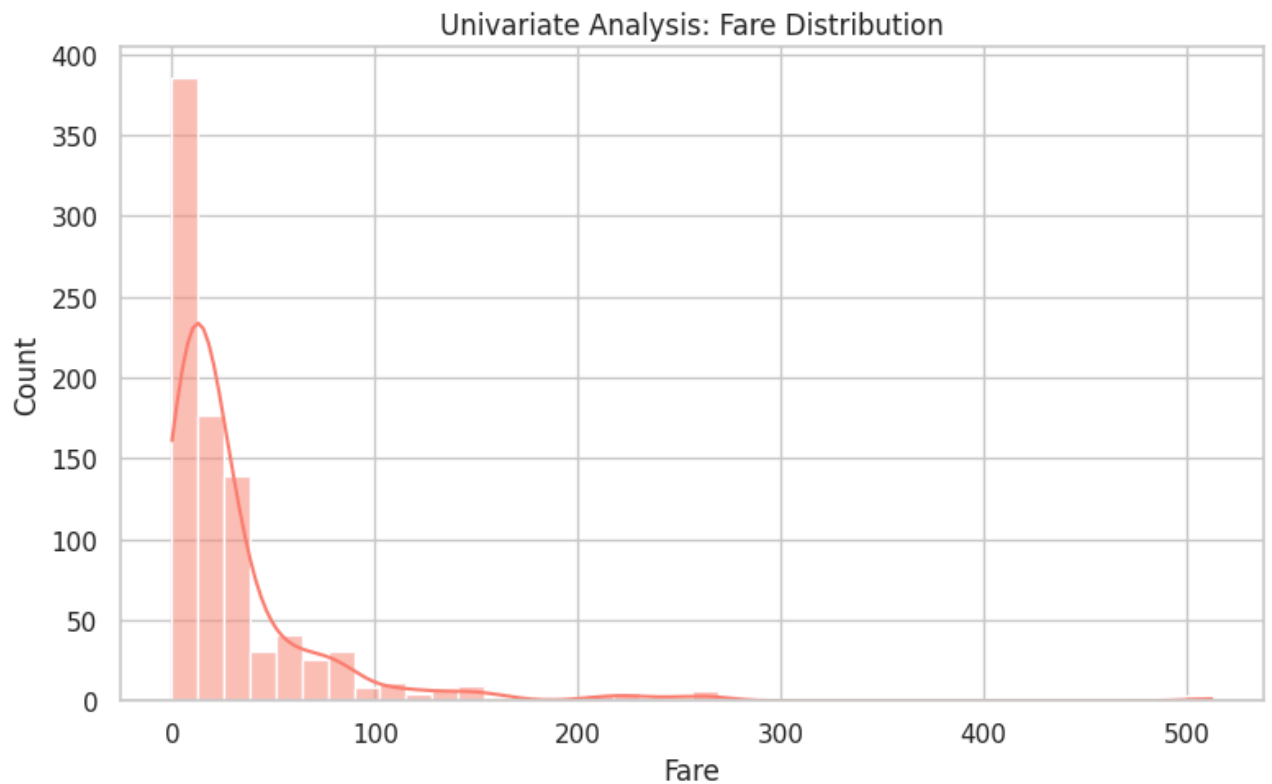
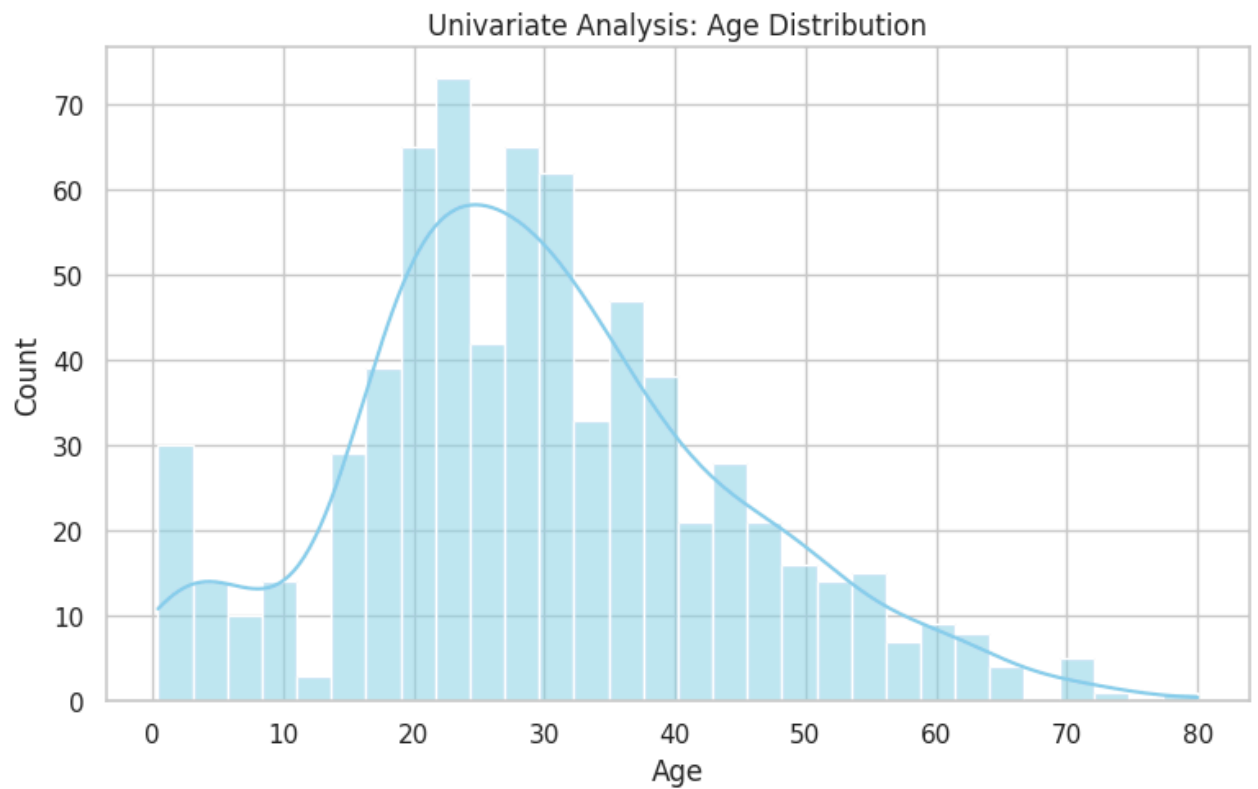
```
In [ ]: # Histograms for Age and Fare distributions

fig, axes = plt.subplots(2, 1, figsize=(8, 10))

sns.histplot(
    df["Age"].dropna(), kde=True, ax=axes[0], color="skyblue", bins=30
)
axes[0].set_title("Univariate Analysis: Age Distribution")
axes[0].set_xlabel("Age")
axes[0].set_ylabel("Count")

sns.histplot(df["Fare"], kde=True, ax=axes[1], color="salmon", bins=40)
axes[1].set_title("Univariate Analysis: Fare Distribution")
axes[1].set_xlabel("Fare")
axes[1].set_ylabel("Count")

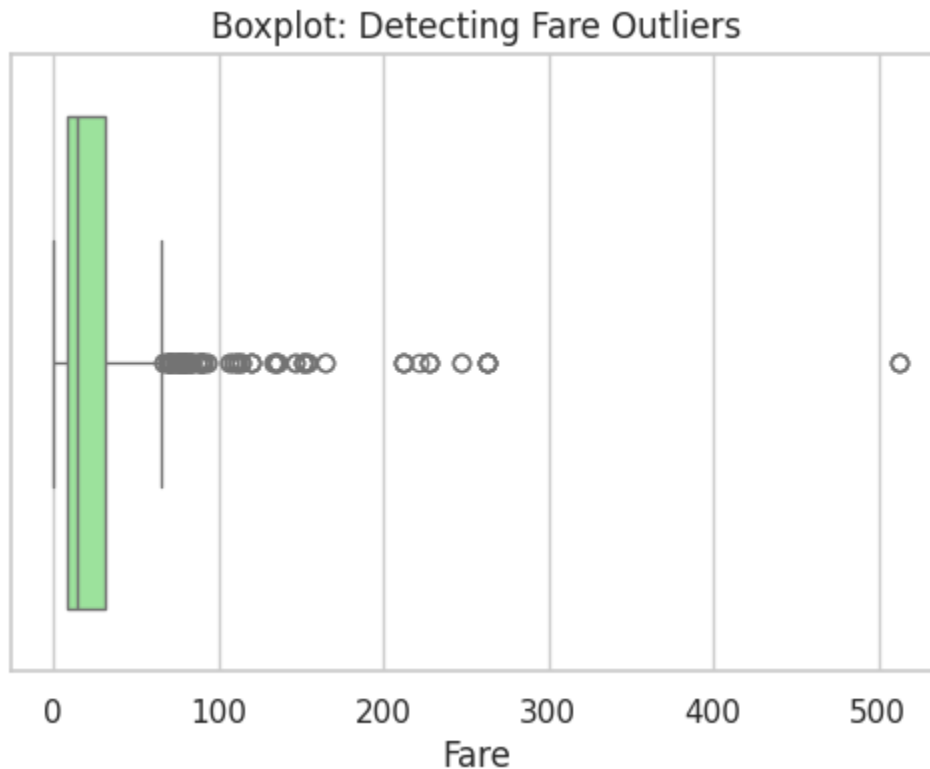
plt.tight_layout()
plt.show()
```



Age Distribution: Passenger ages are roughly bell-shaped and right-skewed, centered around a mean of 29.70 years and a median of 28.00 years, with a notable cluster of infants and young children.

```
In [ ]: # Boxplot
```

```
plt.figure(figsize=(6, 4))
sns.boxplot(x=df["Fare"], color="lightgreen")
plt.title("Boxplot: Detecting Fare Outliers")
plt.xlabel("Fare")
plt.show()
```

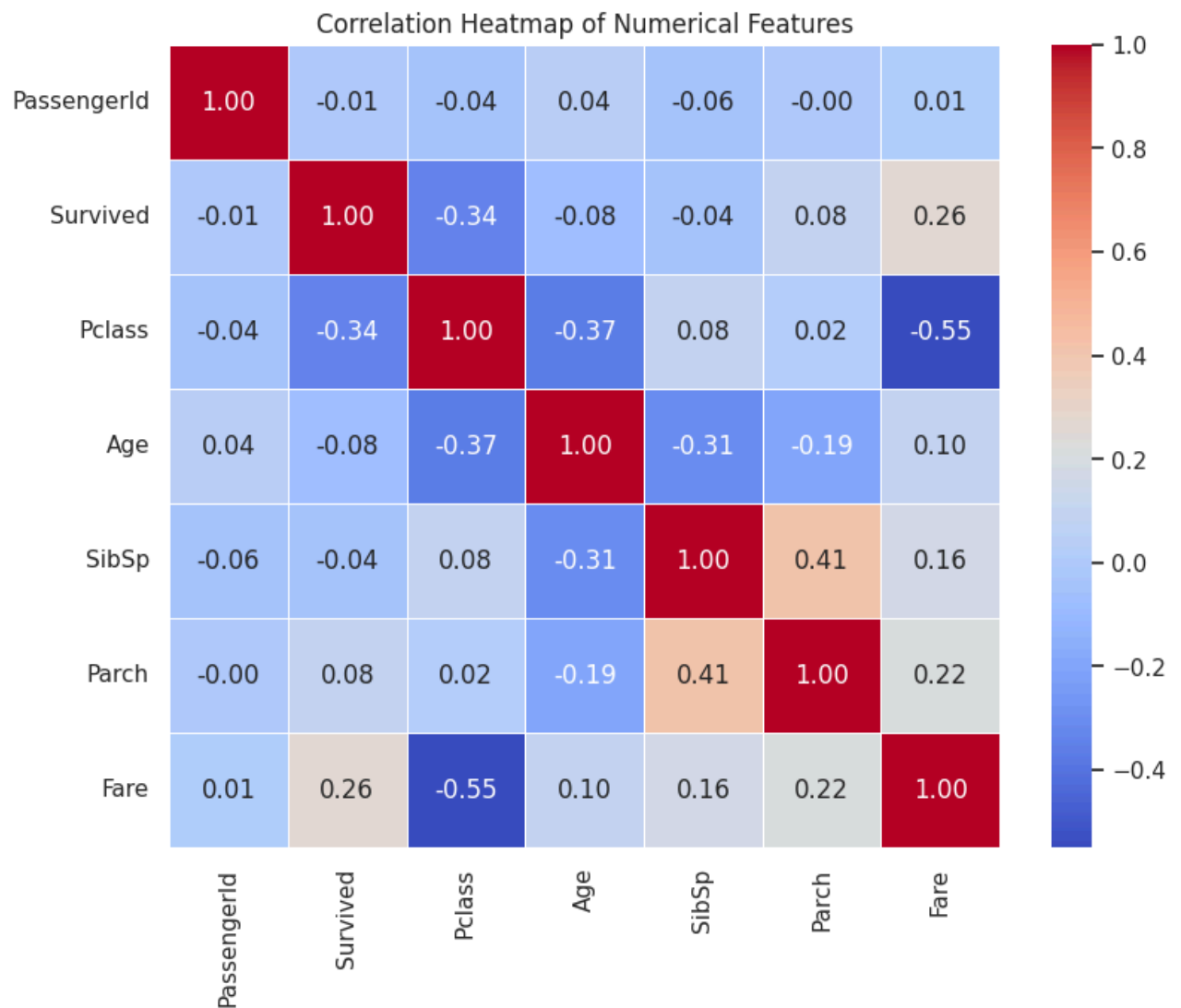


Fare Distribution: Ticket prices are severely right-skewed. While the median fare was \$14.45, the mean fare was \$32.20, driven by extreme high-value outliers paying up to \$512.33.

```
In [ ]: # Correlation Heatmap (Numerical Relationships)
```

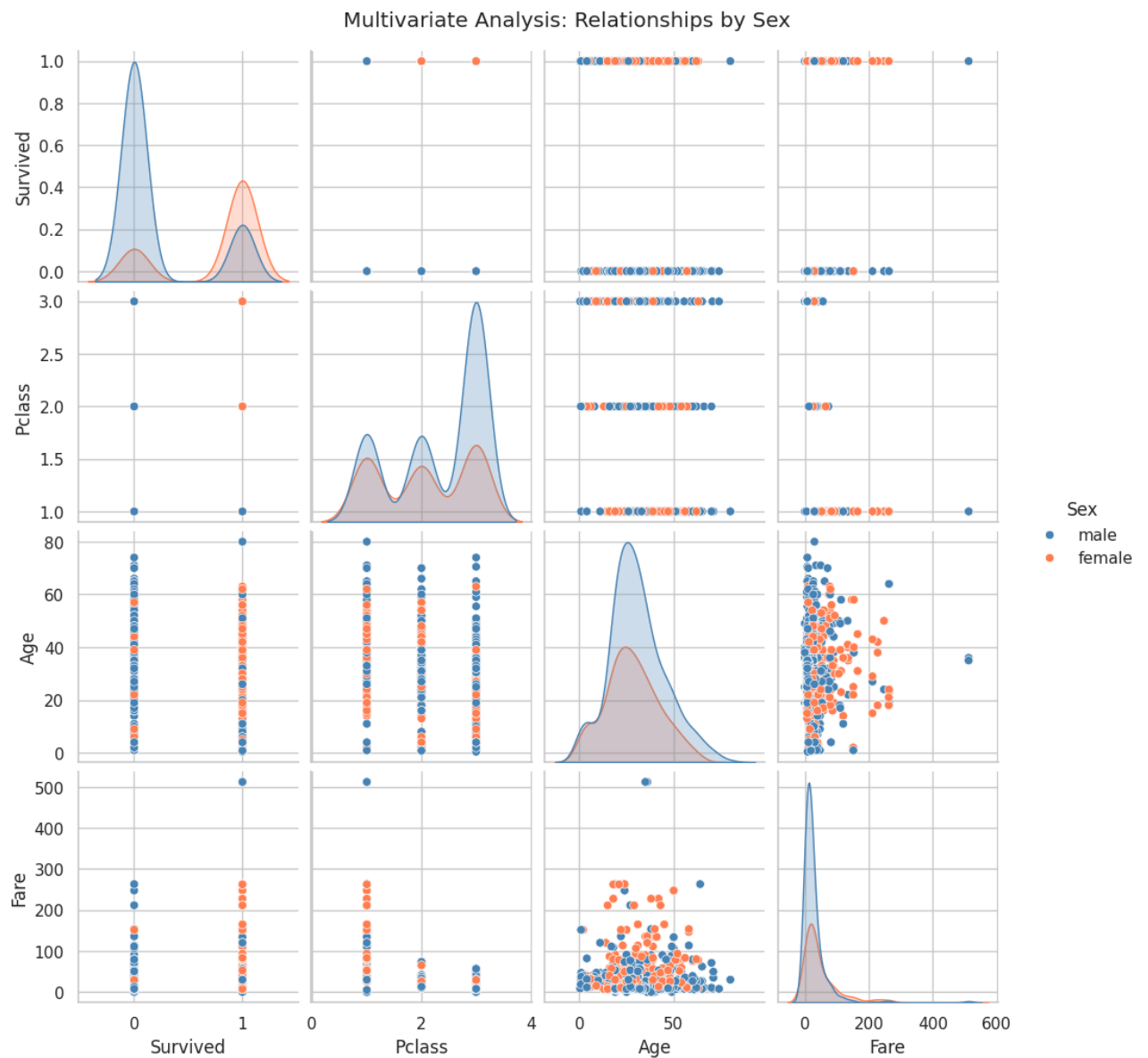
```
plt.figure(figsize=(9, 7))
numeric_df = df.select_dtypes(include=[np.number])
corr_matrix = numeric_df.corr()

sns.heatmap(
    corr_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5
)
plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```



The correlation heatmap revealed that Pclass and Fare had a strong negative correlation, Pclass and Survived had a negative relationship, and Fare and Survived had a positive relationship

```
In [ ]: # Pairplot with Hue for Multivariate Relationships
subset_df = df[["Survived", "Pclass", "Age", "Fare", "Sex"]].dropna()
pair_plot = sns.pairplot(
    subset_df, hue="Sex", palette={"male": "steelblue", "female": "coral"}
)
pair_plot.fig.suptitle(
    "Multivariate Analysis: Relationships by Sex", y=1.02
)
plt.show()
```



The multivariate pairplot showed clear separation by sex across most variables, with females surviving at much higher rates regardless of class or fare